Contents

Parameter Estimation **1**
Bias 1
Estimator Variance 1
Standard Estimators 2

Parameter Estimation

EEE 554 Lecture #13

Consider a family of PDFs indexed by a parameter $\theta \in \mathbb{R}^m$ and denoted $\{f_X(x|\theta)\}$. For the moment, X will be assumed to be a (scalar) RV.

Example: The family is unit-variance normal PDFs indexed by their means $\theta = \mu_X \in \mathbb{R}$:

$$f_X(x|\theta) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(x-\theta)^2\right\}$$

Suppose $X_1, \dots, X_n$ are iid RVs drawn from $f_X(x|\theta)$ for a fixed value of θ. An estimator for the parameter θ from data $\underline{x} = (x_1, \dots, x_n)$ is a function:

$$g : \mathbb{R}^n \to \mathbb{R}^m$$

$$\hat{\theta} = g(X_1, \dots, X_n) = g(\underline{X})$$

Note that the estimate $\hat{\theta}$ is a RV because it is a function of the RVs $X_1, \dots, X_n$. Each realization of $\underline{X}$ gives a realization of $\hat{\theta}$.

Bias

The bias of an estimator $\hat{\theta}$ of θ is:

$$\beta(\hat{\theta}) = E[\hat{\theta}] - \theta$$

An estimator is unbiased if $\beta(\hat{\theta}) = 0$; i.e., if $E[\hat{\theta}] = \theta$.

Example (Sample mean estimator): Suppose $X_1, \dots, X_n$ are iid with mean $\theta = \mu_X$. Define an estimator for θ by:

$$\hat{\theta} = g(X_1, \dots, X_n) = \frac{1}{n}(X_1 + \cdots + X_n)$$

Then:

$$E[\hat{\theta}] = \frac{1}{n}E[X_1 + \cdots + X_n] = \frac{1}{n}(n\mu_X) = \mu_X = \theta$$

So $\hat{\theta}$ is an unbiased estimator of the mean.

Estimator Variance

A second measure of estimator quality is its variance. For an unbiased estimator, small variance ensures that the probability that a realization of $\hat{\theta}$ will be near θ is high. *(Note: The original notes contained a sketch comparing the PDFs of two unbiased estimators, illustrating "large variance" vs "small variance" around θ.)*

Example (cont'd): For the sample mean estimator $\hat{\theta} = \frac{1}{n}(X_1 + \cdots + X_n)$:

$$var(\hat{\theta}) = \frac{1}{n^2} var(X_1 + \cdots + X_n) = \frac{1}{n} var(X)$$

Note that the variance of this unbiased estimator of μ_X decreases as the number n of independent data points increases. An estimator with this property is called **consistent**.

Standard Estimators

- **1. Sample Mean**
  - $$\hat{\mu}_X = \frac{1}{n} \sum_{k=1}^{n} X_k$$
  - Unbiased estimator for $\mu_X = E[X]$.
  - $$var(\hat{\mu}_X) = \frac{1}{n} var(X)$$
- **2. Sample Variance Estimator**
  - $$\hat{v} = \frac{1}{n-1} \sum_{k=1}^{n} (X_k - \hat{\mu}_X)^2$$
  - **Bias:**
  $$\begin{aligned} E[\hat{v}] &= \frac{1}{n-1} E\left[\sum_{k=1}^{n} (X_k^2 - 2X_k\hat{\mu}_X + \hat{\mu}_X^2)\right] \\ &= \frac{1}{n-1} \left\{ \sum_{k=1}^{n} E[X_k^2] - 2E\left[\hat{\mu}_X \sum_{k=1}^{n} X_k\right] + nE[\hat{\mu}_X^2] \right\} \\ &= \frac{1}{n-1} \left\{ nE[X^2] - 2nE[\hat{\mu}_X^2] + nE[\hat{\mu}_X^2] \right\} \\ &= \frac{n}{n-1} \left(E[X^2] - E[\hat{\mu}_X^2] \right) \\ &= \frac{n}{n-1} \left((var(X) + \mu_X^2) - (var(\hat{\mu}_X) + \mu_X^2) \right) \\ &= \frac{n}{n-1} \left(var(X) - \frac{1}{n} var(X) \right) \\ &= \frac{n}{n-1} \left(\frac{n-1}{n} var(X) \right) = var(X) \end{aligned}$$

    Therefore, $\hat{v}$ is an unbiased estimator of $var(X)$.
  - **Variance:** In general, $var(\hat{v})$ is not possible to calculate without additional assumptions on f_X. If X is Gaussian:
  $$var(\hat{v}) = \frac{2(var(X))^2}{n-1}$$
    and $\hat{v}$ is a consistent estimator of $var(X)$.
- **3. Sample Covariance Estimator**
  - With data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ independently drawn from a joint PDF f_{XY}, the sample covariance is:
  - $$\hat{C}_{XY} = \frac{1}{n-1} \sum_{k=1}^{n} (X_k - \hat{\mu}_X)(Y_k - \hat{\mu}_Y)$$
  - This is an unbiased estimator of $cov(X, Y)$.